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Agricultural trench depth systems, methods, and apparatus

Abstract

Apparatus for adjusting the depth of a trench opened by a row unit (10) of an agricultural planter. The row unit (10) includes a trench depth adjustment assembly (90) configured to modify the trench depth. The trench depth adjustment assembly (90) includes a depth adjustment body (94) pivotally connected via a pivot (92) to a frame member (14) of the row unit (10). An electric motor (3030) is operable to cause rotation of a shaft (3034) operably coupled with the depth adjustment body (94), whereby rotation of the shaft (3034) causes the depth adjustment body (94) to pivotally move about the pivot (92) thereby changing a position of contact of the depth adjustment body (94) with a gauge wheel arm (54), thus changing the amount of upward travel of the gauge wheel (52) with respect to a trench opening disc (62) and thus the depth of the trench.

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Background/Summary

BACKGROUND

(1) In recent years, farmers have recognized the need to select and maintain the proper planting depth to ensure the proper seed environment (e.g., temperature and moisture) and seedling emergence. To improve agronomic practices, it would also be desirable for the farmer to understand the relationship between actual planting depth and metrics such as emergence and yield.

Conventional agricultural planters include only apparatus for adjusting a maximum planting depth, which may not be maintained during operation due to soil conditions or insufficient down-pressure on the planter row unit. Even in operation of modern planters having sensors for determining whether full trench depth has been lost, the actual depth planted is still not determined. Thus there is a need for systems, methods and apparatus for controlling and/or measuring the depth of a trench opened by an agricultural planter.

Description

DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a right side elevation view of an embodiment of an agricultural row unit.
- (2) FIG. 2 is a right side elevation view of another embodiment of an agricultural row unit with certain components removed for clarity.
- (3) FIG. 3 is a perspective view of the agricultural row unit of FIG. 2.
- (4) FIG. 4 is a perspective view of the agricultural row unit of FIG. 2 with a right gauge wheel removed for clarity.
- (5) FIG. 5 is an enlarged partial right side elevation view of the agricultural row unit of FIG. 2.
- (6) FIG. 6 is a rear elevation view of the agricultural row unit of FIG. 2.
- (7) FIG. 7 is a side elevation view of an embodiment of a depth adjustment assembly and a secondary depth adjustment assembly.
- (8) FIG. 8 is a side elevation view of another embodiment of a depth adjustment assembly and a secondary depth adjustment assembly.
- (9) FIG. 9 is a side elevation view of another embodiment of a depth adjustment assembly and a secondary depth adjustment assembly.
- (10) FIG. 10 is a side elevation view of another embodiment of a depth adjustment assembly and a

secondary depth adjustment assembly.

(11) FIG. 10A is a side elevation view of another embodiment of a depth adjustment assembly and a secondary depth adjustment assembly.

(12) FIG. 11 schematically illustrates an embodiment of a system for controlling furrow depth.

(13) FIG. 12 is a side elevation view of another embodiment of a depth adjustment assembly and a secondary depth adjustment assembly.

(14) FIG. 13 is a perspective view of another embodiment of a depth adjustment assembly and a secondary depth adjustment assembly disposed on the row unit frame.

(15) FIG. 13A is a side elevation view of the depth adjustment assembly and a secondary depth adjustment assembly of FIG. 13 as viewed along lines X-X of FIG. 13.

(16) FIG. 13B is an enlarged perspective view of the depth adjustment assembly and a secondary adjustment assembly of FIG. 13 with the row unit frame removed.

(17) FIG. 14 is a perspective view of another embodiment of a depth adjustment assembly and a secondary depth adjustment assembly disposed on the row unit frame.

(18) FIG. 14A is a side elevation view of the depth adjustment assembly and a secondary depth adjustment assembly of FIG. 14 as viewed along lines Y-Y of FIG. 14.

(19) FIG. 14B is a side elevation view of the depth adjustment assembly and a secondary depth adjustment assembly of FIG. 14 showing an alternative embodiment in which the roller replaced with a cog.

(20) FIG. 15 is a perspective view of another embodiment of a depth adjustment assembly with a rotary actuator disposed on the row unit frame.

(21) FIG. 15A is a side elevation view of the depth adjustment assembly of FIG. 15.

(22) FIG. 15B is a side elevation view of the depth adjustment assembly of FIG. 15A including a manual adjustment.

(23) FIG. 16 is a partial perspective view of another embodiment of a depth adjustment assembly with a rotary actuator disposed on a gear rack on the row unit frame.

(24) FIG. 16A is a side elevation and partial cutaway view of the depth adjustment assembly of FIG. 16.

(25) FIG. 16B is a side elevation and partial cutaway view of another embodiment for the depth adjustment assembly of FIG. 16.

(26) FIG. 16C is a side elevation and partial cutaway view of another embodiment for the depth adjustment assembly of FIG. 16

(27) FIG. 16D is a rear elevation view of the embodiment of FIG. 16C.

(28) FIG. 16E is a rear elevation view of another embodiment for a depth adjustment assembly.

(29) FIG. 17 is a side elevation view showing a Case row unit adapted with another embodiment of a depth adjustment assembly.

(30) FIG. 17A is an enlarged view of the embodiment of FIG. 17.

(31) FIG. 18 is a side elevation and partial cutaway view of another embodiment of a depth adjustment assembly with a position location system.

(32) FIG. 18A is rear elevation view of the embodiment of FIG. 18.

(33) FIG. 19 is a side elevation and partial cutaway view of another embodiment of a depth adjustment assembly with a position location system.

(34) FIG. 19A is rear elevation view of the embodiment of FIG. 19.

(35) FIG. 20A is a perspective view of another embodiment of a depth adjustment assembly with a position location system shown mounted to a row unit frame member.

(36) FIG. 20B is an enlarged perspective view of the depth adjustment assembly embodiment of FIG. 20A with the row unit frame member removed.

(37) FIG. 20C is a perspective view of the embodiment of FIG. 20B with the gear rack removed.

(38) FIG. 20D is a right side view of the embodiment of FIG. 20B.

(39) FIG. 20E is a right side view of the embodiment of FIG. 20C.

- (40) FIG. 20F is a rear view of the embodiment of FIG. 20C.
- (41) FIG. 20G is a perspective view of the bottom of the gear rack of the embodiment of FIG.
- (42) FIG. 20H is another perspective view of the gear rack and rollers of the embodiment of FIG. 20A.
- (43) FIG. 20I is a another perspective view of the gear rack of FIG. 20A.
- (44) FIG. 20J is a perspective view of the gear box of the embodiment of FIG. 20A.
- (45) FIG. 20K is a perspective view of the worm gear and wheel inside of the gear box of FIG. 20J.
- (46) FIG. 21 is a side elevation and partial cutaway view of another embodiment of a depth adjustment assembly with a position location system.
- (47) FIG. 22 is a rear perspective view of another embodiment of a depth adjustment assembly similar to the embodiment of FIG. 20A and including a position location system.
- (48) FIG. 22A is a front perspective view the embodiment of the depth adjustment assembly of FIG. 22.
- (49) FIGS. 23A and 23B are enlarged perspective views of the embodiment of FIG. 22 showing the UHMVV guides.
- (50) FIG. 24 is an enlarged perspective view of the embodiment of FIG. 22 showing the cover removed from the motor and circuit board to show the magnet and a Hall effect sensor and current sensor on the circuit board.
- (51) FIG. 25 is another enlarged perspective view of the embodiment of FIG. 22 showing the cover removed from the motor and circuit board to show the magnet, the Hall effect sensor, another current sensor and an accelerometer on the circuit board.
- (52) FIG. 26 is an enlarged cross-sectional view of the embodiment of FIG. 22 as viewed along lines 26-26 of FIG. 22, showing the magnet in relation to the Hall effect sensor on the circuit board within the cover.
- (53) FIG. 27 is a rear perspective view of another embodiment of a depth adjustment assembly similar to the embodiment of FIG. 22 and incorporating a homing system.
- (54) FIG. 28 is a partial perspective view of another embodiment of a depth adjustment assembly shown mounted to a frame member of a planter row unit.
- (55) FIG. 29 is the same perspective view of the depth adjustment assembly of FIG. 28, but with components removed to show the internal components of an embodiment of the gear box.
- (56) FIG. 30 is another perspective view of the depth adjustment assembly shown in FIG. 29 as viewed from the forward end looking rearward.
- (57) FIG. 31 is a side elevation view of the depth adjustment assembly as shown in FIG. 29 and illustrating the pivotal movement of the depth adjustment body.
- (58) FIG. 32 is a partial perspective view of another embodiment of a depth adjustment assembly shown mounted to a frame member of a planter row unit.
- (59) FIG. 33 is the same perspective view of the depth adjustment assembly of FIG. 32, but with components removed to show the internal components of an embodiment of the gear box.
- (60) FIG. 34 is another perspective view of the depth adjustment assembly shown in FIG. 33 as viewed from the forward end looking rearward.
- (61) FIG. 35 is a side elevation view of the depth adjustment assembly as shown in FIG. 33 and illustrating the pivotal movement of the depth adjustment body.
- (62) FIG. 36 is a partial perspective view of another embodiment of a depth adjustment assembly.
- (63) FIG. 37 is another perspective view of the depth adjustment assembly of FIG. 36 as viewed from the forward end looking rearward, but with components removed to show the internal components of an embodiment of the gear box.
- (64) FIG. 38 is a side elevation view of the depth adjustment assembly as shown in FIG. 37 and illustrating the pivotal movement of the depth adjustment body.
- (65) FIG. 39 is a partial perspective view of another embodiment of a depth adjustment assembly.
- (66) FIG. 40 is another perspective view of the depth adjustment assembly of FIG. 38 as viewed

from the forward end looking rearward, but with components removed to show the internal components of an embodiment of the gear box.

(67) FIG. **41** is a side elevation view of the depth adjustment assembly as shown in FIG. **40** and illustrating the pivotal movement of the depth adjustment body.

(68) FIG. **42** is a perspective view showing an embodiment of a motor and gearbox of an actuator for a depth adjustment assembly which incorporates an inductance sensor.

(69) FIG. **43** is a cross-sectional perspective view of FIG. **42** as viewed along lines **43-43** of FIG. **42** and showing an embodiment of an inductance sensor.

(70) FIG. **44** is an elevation view of FIG. **43** showing the inductance sensor with the gearbox housing shown in phantom lines for clarity.

(71) FIG. **45** is an enlarged perspective view of the gear wheel shown in FIG. **42** and showing an embodiment of an interchangeable cam for providing different targets with different eccentric curved surface profiles.

(72) FIG. **46** is an exploded view of FIG. **45** showing the interchangeable cam removed from the gear wheel.

(73) FIG. **47** is another perspective view of the interchangeable cam of FIG. **46**.

DESCRIPTION

(74) Referring now to the drawings, wherein like reference numerals designate identical or corresponding parts throughout the several views, FIG. **1** illustrates an agricultural implement, e.g., a planter, comprising a toolbar **8** to which multiple row units **10** are mounted in transversely spaced relation. Each row unit **10** is preferably mounted to the toolbar by a parallel arm arrangement **16** such that the row unit is permitted to translate vertically with respect to the toolbar. A downforce actuator **18** may be mounted to the toolbar **8** and the parallel arm arrangement **16** and configured to apply supplemental down-pressure to the row unit **10**.

(75) The row unit **10** includes a frame **14** which supports an opening disc assembly **60**. The opening assembly **60** may include two angled opening discs **62** rollingly mounted to a downwardly extending shank **15** of the frame **14**. The opening discs **62** open a v-shaped trench **3** (i.e., furrow, seed furrow) in the soil surface **7** as the row unit **10** advances in a forward direction of travel through the field. The row unit **10** includes a gauge wheel assembly **50** which may comprise two gauge wheels **52** pivotally mounted to either side of the frame **14** by two gauge wheel arms **54**. The gauge wheels **52** roll along the surface of the soil. A depth adjustment assembly **90** pivotally mounted to the frame **14** at a pivot **92** contacts the gauge wheel arms **54** to limit the upward travel of the gauge wheel arms **54**, thus limiting the depth of the trench opened by the opening disc assembly **60**. A closing assembly **40** is preferably pivotally coupled to the frame **14** and configured to move soil back into the trench **3**.

(76) Continuing to refer to FIG. **1**, seeds **5** are communicated from a hopper **12** to a seed meter **30** configured to singulate the supplied seeds. The meter **30** may be a vacuum-type meter such as that disclosed in Applicant's International Patent Pub. No. WO/2012/129442, the disclosure of which is hereby incorporated by reference herein in its entirety. In operation, the seed meter **30** preferably deposits the supplied seeds into a seed tube **32** which may be removably mounted to the frame **14**. In operation, seeds **5** deposited by the meter **30** fall through the seed tube **32** into the trench **3**.

(77) Turning to FIGS. **2-6**, the depth adjustment assembly **90** is illustrated in more detail. The depth adjustment assembly **90** includes a rocker **95** pivotally mounted to a depth adjustment body **94**. The depth adjustment body **94** is pivotally mounted to the row unit frame **14** about the pivot **92**. A handle **98** is slidably received within the depth adjustment body **94** such that the user can selectively engage and disengage the handle (e.g., left and right hooks **99-1**, **99-2**, respectively, which may be formed as a part of the handle **98**) with one of a plurality of depth adjustment slots **97** (FIG. **6**) formed within the row unit frame **14**. With reference to FIG. **7**, the handle **98** is partially slidingly received within a cavity **710** of the depth adjustment body **94**, and an optional spring **730** engages an annular lip **740** disposed on the bottom end of the handle **98**. The spring **730**

imposes a resilient force to retain the hooks **99** (FIG. 6) in the selected slot **97**, but permits the user to withdraw the handle **98** to temporarily disengage the hooks **99** from the slot **97**. In operation, the upward travel of the gauge wheels **52** is limited by contact of the gauge wheel arms **54** with the rocker **95**. When one of the gauge wheels, e.g., left gauge wheel **52-1**, encounters an obstruction, the rocker **95** allows the left gauge wheel arm **54-1** to travel upward while lowering the right gauge wheel **52-2** by the same absolute displacement such that the row unit **10** rises by half the height of the obstruction. It should be appreciated that there are other row units with manual adjustments similar to those described herein. Non-limiting examples can be found in U.S. Publication Nos. US20170000003 and US20170006757, both of which are incorporated herein by reference. The depth adjustment assemblies described herein work with similar systems with rockers, pivot, and adjustment arms.

(78) It should be appreciated that the handle **98** and depth adjustment body **94** comprise a primary depth adjustment sub-assembly configured to permit the user to select one of a plurality of pre-selected furrow depths. The pre-selected furrow depths each correspond to one of the depth adjustment slots **97**. In some embodiments, as described in detail later, rather than using the handle **98** to manually select a depth adjustment slot, an actuator may be used to adjust the position of handle **98**; for example, a linear actuator (not shown) mounted to the row unit frame **14** may be disposed to adjust the position of the handle **98**. Alternatively, a rotary actuator may turn a gear which adjusts the position of the handle relative to the depth adjustment slots **97**.

(79) In each of the embodiments illustrated in FIGS. 7-10A and 12, a secondary depth adjustment assembly is configured to modify one or more of the pre-selected furrow depths. The secondary depth adjustment assembly may modify the pre-selected furrow depths by more precise adjustments (e.g., by smaller adjustment steps) than the depth modifications enabled by the primary depth adjustment assembly (e.g., by selecting which depth adjustment slot **97** is engaged by the handle **98**). For example, referring FIG. 7, the depth adjustment assembly **90A** includes an actuator **720** which adjusts the effective length of the depth adjustment assembly **90A**. In the illustrated embodiment, the extension of actuator **720** determines the position of the rocker **95** relative to the depth adjustment body **94**. As illustrated, the rocker **95** is pivotally mounted to a movable member **770** having a cavity **775** for receiving a protrusion **760** mounted to (or formed as a part with) the depth adjustment body **94**. The protrusion **760** and cavity **775** retain the alignment of the moveable member relative to the depth adjustment body **94**, but permit the actuator **720** to modify the position along an axis which is parallel to the pivot axis of the rocker **95**. It should be appreciated that modification of the extension of actuator **720** (and thus the effective length of the depth adjustment assembly) modifies the furrow depth for any given depth setting of the handle **98**. Any of the secondary depth adjustment assemblies described herein may be used as the only depth adjustment, such that the primary depth adjustment does not need to be set, and whereby the secondary depth adjustment can adjust the depth adjustment body **94** across the entire range of depth settings.

(80) FIG. 8 illustrates another embodiment of a depth adjustment assembly **90B** having a secondary depth adjustment assembly wherein an actuator **800** modifies the angular position at which one or more gauge wheel arms **54** is stopped by the depth adjustment assembly **90B** for any given setting of the depth adjustment handle **98**. The actuator **800** adjusts a position of a surface **810** which is pivotally mounted to the gauge wheel arm **54**; the surface **810** is disposed to contact the rocker **95** at the point of maximum upward travel of the gauge wheel arm **54**. Extension of the actuator **800** and thus modification of the position of surface **810** thus modifies the point of maximum upward travel of the gauge wheel and thus modifies the furrow depth determined by the gauge wheel. In some embodiments, a functionally similar actuator **800** and pivotally-mounted surface **810** may be mounted to both gauge wheel arms **54**.

(81) FIG. 9 illustrates another embodiment of a depth adjustment assembly **90C** having a secondary depth adjustment assembly wherein a modified rocker **900** is configured to modify its shape in

order to modify the furrow depth for any given depth setting of the handle **98**. The rocker **900** includes portions **910-1**, **910-2** which contact the gauge wheel arms **54-1** and **54-2**, respectively, to limit the upward travel of the gauge wheel arms. An actuator **950** changes the angle between the portions **910-1** and **910-2** and thus the shape of the rocker **900**. Retraction of the actuator **950** raises the members **910** and thus modifies the maximum height of the gauge wheel arms **54** and the furrow depth.

(82) FIG. **10** illustrates another embodiment of a depth adjustment assembly **90D** having a secondary depth adjustment assembly wherein the rocker **95** is pivotally mounted to the depth adjustment body **94**, preferably about a laterally extending axis defined by pivot **1010**. An actuator **1000** preferably determines the angular position of the rocker **95** about the pivot **1010** relative to the depth adjustment body **94**, thus modifying the maximum upward travel of the gauge wheel arms **54** and the furrow depth.

(83) FIG. **10A** illustrates an alternative to the embodiment illustrated in FIG. **10**. In this embodiment, pivot **1010** from FIG. **10** is removed, and rocker **95** is attached to connector **1011**, which pivots about pivot **92**.

(84) FIG. **12** illustrates yet another embodiment of a depth adjustment assembly **90E** having a secondary depth adjustment assembly wherein an actuator **1230** advances a depth adjustment member **1210** (e.g., a wedge) which is slidably fixed to the gauge wheel arm and disposed to slide along the length of the gauge wheel arm **54**. An actuator **1230** (e.g., a linear actuator such as an electric, hydraulic, or pneumatic actuator) selectively modifies (e.g., by extension or retraction) the position of the depth adjustment member **1210**, e.g., along the length of the gauge wheel arm **54**. The position of the depth adjustment member **1210** along the length of the gauge wheel arm modifies the uppermost angular position of the gauge wheel arm relative to the rocker and thus modifies the depth of the furrow opened by the row unit in operation. The actuator **1230** may be mounted to the gauge wheel arm **54**, e.g., by being fixed to a plate **1225** mounted to the gauge wheel arm **54**.

(85) In some embodiments, the actuator **1230** may adjust the position of the depth adjustment member **1210** by means of a biasing mechanism. The biasing mechanism may increase or reduce a biasing force on the wedge **1210** as the actuator **1230** is extended. For example, as illustrated in FIG. **12**, the actuator **1230** may modify a position of a biasing member such as a plate **1220** relative to the depth adjustment member **1210**. Optionally, a first spring **1215a** may be fixed to the depth adjustment member **1210** at a first end thereof and may be fixed to the plate **1220** at a second end thereof. Optionally, a second spring **1215b** may be fixed to the plate **1220** at a first end thereof and may be fixed to the plate **1225** at a second end thereof. In the undeflected position shown in FIG. **12**, neither of the springs **1215a**, **1215b** impose a substantial force on the biasing member **1210**. As the actuator **1230** advances from the undeflected position, the spring imposes an increasing advancing force on the biasing member **1220** (e.g., generally toward the rocker **95**). As the actuator **1230** retracts from the undeflected position, the spring imposes an increasing retracting force on the biasing member **1220** (e.g., generally away from the rocker **95**).

(86) In operation, when a component of force transmitted from the actuator **1230** (e.g., via the spring **1215a** of the biasing mechanism illustrated in FIG. **12**) to the rocker **95** exceeds an oppositely acting force of the rocker **95** on the gauge wheel arm (or on the depth adjustment member if the rocker is already contacting the depth adjustment member), the depth adjustment member **1210** advances, forcing the rocker **95** farther away from the gauge wheel arm and reducing the furrow depth. It should be appreciated that the biasing force may be built up gradually by extension of the actuator **1230** without being sufficient to advance the depth adjustment member **1210** until sufficient extension of the actuator or until reduction of downforce.

(87) FIGS. **13** and **14** are perspective views of a row unit frame **14** showing alternative embodiments of depth adjustment assemblies **90F** and **90G**, respectively, disposed on the row unit **14**.

(88) Referring to FIG. 13A, a side elevation view of depth adjustment assembly 90F is shown as viewed along lines X-X of FIG. 13. FIG. 13B is an enlarged perspective view of depth adjustment assembly 90F with the row unit frame 14 removed and the handle 98 shown in dashed lines for clarity.

(89) The depth adjustment assembly 90F includes a housing 1494 which is received between the sidewalls of the row unit frame 14. The housing 1494 is adjustably positionable along the depth adjustment slots 97 of the row unit frame 14 by engagement of the handle 98 within one of the plurality of depth adjustment slots 97 to achieve the initial preselected furrow depth. The handle 98 includes hooks 99-1, 99-2 which extend into the slots 97, thereby positioning the housing 1494 at the desired slot 97.

(90) The secondary depth adjustment assembly of the depth adjustment assembly 90F comprises an actuator 1450 (such as an electric motor), drive screw 1410, drive member 1420, cam arm 1460 and cog 1430, all of which cooperate to adjustably position the rocker 95 with respect to the row unit frame 14 as hereinafter described.

(91) As shown in FIG. 13A, the drive screw 1410 extends into the housing 1494 and is driven by the actuator 1450. The drive screw 1410 is threadably received by the drive member 1420. The cog 1430 is rotatably disposed on drive member 1420. A cam arm 1460 has a proximal end 1461 and a distal end 1462. The distal end 1462 of the cam arm 1460 is pivotably mounted about pivot 92. The proximal end 1461 of the cam arm 1460 includes teeth 1463 that engage with the cog 1430. The rocker 95 is pivotally attached to the distal end 1462 of the cam arm 1460. Stops 1470-1 and 1470-2 may be disposed in the housing 1494 on either side of cam arm 1460 to limit the rotational movement of cam arm 1460 in both the clockwise and counterclockwise rotation.

(92) In operation, the actuator 1450 rotates the drive screw 1410 causing the drive member 1420 threadably attached thereto to be threaded upwardly or downwardly along the drive screw 1410 such that it is raised and lowered within the housing 1494. If the drive screw 1410 is rotated by the actuator 1450 in the direction to cause the drive member 1420 to be threaded upwardly along the drive screw 1410, the cog 1430 engages with the teeth 1463 of the cam arm 1460 causing the cam arm 1460 to pivot counterclockwise (as shown in FIG. 13A) about pivot 92, which raises the rocker 95 with respect to the row unit frame 14, permitting the gauge wheel arms 54 to raise with respect to the frame member 14, thereby increasing the furrow depth. Conversely, if the drive screw 1410 is rotated by the actuator 1450 in the opposite direction to cause the drive member 1420 to be threaded downwardly along the drive screw 1410, the cog 1430 engages with the teeth 1463 of the cam arm 1460 causing the cam arm 1460 to pivot clockwise (as shown in FIG. 13A) about pivot 92, which forces the rocker 95 lower with respect to the frame member 14, thereby forcing the gauge wheel arms 54 downwardly with respect to the frame member 14 and, in turn, decreasing the furrow depth.

(93) Referring to FIG. 14A, a side elevation view of depth adjustment assembly 90G is shown as viewed along lines Y-Y of FIG. 14. Similar to the embodiment of 90F, the depth adjustment assembly 90G includes a housing 1594 which is received between the sidewalls of the row unit frame 14. The housing 1594 is adjustably positionable along the depth adjustment slots 97 of the row unit frame 14 by engagement of the handle 98 within one of the plurality of depth adjustment slots 97 to achieve the initial preselected furrow depth. The handle 98 includes hooks or pegs 99-1, 99-2 which extend into the slots 97 thereby securing the housing 1594 at the desired slot 97.

(94) The secondary depth adjustment assembly of the depth adjustment assembly 90G comprises an actuator 1550 (such as an electric motor), drive screw 1510, drive member 1520, cam arm 1560 and a roller 1565 (FIG. 14A) or a cog 1530 (FIG. 14B), which cooperate to adjustably position the rocker 95 with respect to the row unit frame 14 as hereinafter described.

(95) As shown in FIG. 14A, the drive screw 1510 extends into the housing 1594 and is driven by the actuator 1550. The drive screw 1510 is threadably received by drive member 1520. The drive member 1520 has a sloped side 1521 that engages with a roller 1565 rotatably attached to a

proximal end **1561** of the cam arm **1560**. A distal end **1562** of the cam arm **1560** is pivotably mounted about pivot **92**. The rocker **95** is pivotally attached to the distal end **1562** of the cam arm **1560**. In an alternative embodiment shown in FIG. **14B**, roller **1565** is replaced with a rotatable cog **1530** and the sloped side **1521** includes teeth **1563** which engage with the cog **1530** as the cog **1530** rotates. Stops **1570-1** and **1570-2** may be disposed in the housing **1594** on either side of cam arm **1560** to limit the rotational movement of cam arm **1560** in both the clockwise and counterclockwise rotation.

(96) In operation, the actuator **1550** rotates the drive screw **1510** causing the drive member **1520** threadably attached thereto to be threaded upwardly or downwardly along the drive screw **1510** such that it is raised and lowered within the housing **1594**. If the drive screw **1510** is rotated by the actuator **1550** in the direction to cause the drive member **1520** to be threaded upwardly along the drive screw **1510**, the roller **1565** will roll downwardly along the sloped side **1521** causing the cam arm **1560** to pivot counterclockwise (as shown in FIG. **14A**) about pivot **92**, which raises the rocker **95** with respect to the row unit frame **14**, permitting the gauge wheel arms **54** to raise with respect to the frame member **14**, thereby increasing the furrow depth. Conversely, if the drive screw **1510** is rotated by the actuator **1550** in the opposite direction to cause the drive member **1520** to be threaded downwardly along the drive screw **1510**, the roller **1565** will roll along the curved surface **1521** causing the cam arm **1560** to pivot clockwise (as shown in FIG. **14A**) about pivot **92**, which forces the rocker **95** lower with respect to the frame member **14**, thereby forcing the gauge wheel arms **54** downwardly with respect to the frame member **14** and, in turn, decreasing the furrow depth. It should be appreciated that with respect to the embodiment shown in FIG. **14B**, wherein the roller **1565** and sloped surface **1521** are replaced with the cog **1530** which engage teeth **1563** on the sloped surface **1521**, the same action is accomplished.

(97) In an alternative embodiment to any of embodiments **90A**, **90B**, **90C**, **90D**, **90E**, **90F**, and the depth adjustment body **94**, **1494**, or **1594** does not need to be adjustable. Depth adjustment body **94**, **1494**, or **1594** can remain fixed with respect to frame **14** and the secondary adjustment assembly of any of embodiments **90A**, **90B**, **90C**, **90D**, **90E**, **90F**, and **90G** will provide the entire range of depth adjustment. Instead of pivoting at pivot **92**, depth adjustment body **94**, **1494** or **1594** is fixed to frame **14**.

(98) Any of the actuators (**720**, **800**, **950**, **1000**, **1230**, **1450**, **1550**) may be electrical, hydraulic, or pneumatic actuators.

(99) FIGS. **15** and **15A** illustrate another embodiment of a depth adjustment assembly **90H** in which a rotary actuator **1650** (such as an electric motor) turns gears **1640-1** and **1640-2** that adjust the position of the depth adjustment body **1694** relative to the depth adjustment slots **97**. Gears **1640-1** and **1640-2** have teeth **1641-1** and **1641-2**, respectively that engage in slots **97**. Rotary actuator **1650** is connected to depth adjustment body **1694**, which is pivotally mounted to the frame **14** at pivot **92**. Rocker **95** is pivotally mounted to the depth adjustment body **1694**. Rotary actuator may be gear reduced (such as 300:1) to allow for smaller rotation of gears **1640-1** and **1640-2**. In this embodiment, rotary actuator **1650** replaces handle **98**. This embodiment can be used as the only depth adjustment assembly, or it can be used as the primary depth adjustment assembly and used in combination with any of the other secondary depth adjustment assemblies previously described.

(100) FIG. **15B** illustrates an alternative embodiment of a depth adjustment assembly **90H-1** similar to depth adjustment assembly **90H**, but in which depth adjustment body **1694** is replaced with depth adjustment body **1695**, handle shaft **1698**, and spring **1630**. Handle shaft **1698** is attached to actuator **1650** and is partially slidably received within a cavity **1696** of the depth adjustment body **1695**. The spring **1630** engages an annular lip **1680** disposed on the bottom end of the handle shaft **1698**. The spring **1630** thus imposes a resilient force to retain the gears **1640** in the selected slot **97**, but permits the user to withdraw the actuator **1650** using handle **1660** attached to actuator **1650** to temporarily disengage the gears **1640** from the slot **97** to a desired pre-set depth to

minimize the amount of travel that the actuator **1650** needs to reach a selected depth.

(101) FIGS. **16** and **16A** illustrate another embodiment of a depth adjustment assembly **90I** in which a gear rack **1710** is disposed on row unit **14** over depth adjustment slots **97**. A radius **R** (FIG. **16A**) from pivot **92** to gear rack **1710** remains constant along the gear rack **1710** having two rows of teeth **1716-1**, **1716-2**. Rotary actuator **1750** is disposed over gear rack **1710** and is connected to a handle shaft **1798** at gear box **1720**. Rotary actuator **1750** includes a motor **1730** connected to a gear box **1720**. In the rear perspective view of FIG. **16**, the rotary actuator **1750** is removed for clarity to better show the gear rack **1710**. Gear box **1720** has gears **1740** having teeth **1741** for meshing with gear rack **1710**. Only one of the gears is visible in FIG. **16A**, but it should be appreciated that respective gears **1740-1**, **1740-2**, having respective teeth **1741-1**, **1741-2** would rotatably engage with respective teeth **1716-1**, **1716-2** of gear rack **1710**. A handle **1799** can be disposed on motor **1730** to permit rotary actuator **1750** to disengage from gear rack **1710** for moving to a different position on gear rack **1710** to preset a selected depth. Rotary actuator **1750** may be gear reduced (such as 300:1) to allow for smaller rotation of gears **1740-1** and **1740-2**. In this embodiment, rotary actuator **1750** replaces handle **98** described in the previous embodiments. Handle shaft **1798** is attached to actuator **1750** at gear box **1720** and is partially slidably received within a cavity **1796** of a depth adjustment body **1794**. A spring **1791** engages an annular lip **1795** disposed on the bottom end of the handle shaft **1798**. The spring **1791** imposes a resilient force to retain the gears **1740** meshed with gear rack **1710** but permits the user to withdraw the actuator **1750** using handle **1799** attached to actuator **1750** to temporarily disengage the gears **1740** from gear rack **1710**. Depth adjustment body **1794** is pivotally mounted to the frame **14** at pivot **92**. Rocker **95** is pivotally mounted to the depth adjustment body **1794**.

(102) FIG. **16B** illustrates an alternative embodiment of a depth adjustment assembly **90I-1**, similar to the depth adjustment assembly **90I**, but in which handle **1799** is replaced with manual adjustment **1780**. Manual adjustment **1780** may be a knob, a bolt head or other suitable means to permit a user to manually move motor **1730** by hand or a tool to adjust depth adjustment assembly **90I-1** when motor **1730** cannot be driven electrically.

(103) FIG. **16C** is a side elevation and partial cutaway view of another embodiment of a depth adjustment assembly **90J** that further includes a rotary actuator **1750A**. FIG. **16D** is a rear elevation view of the embodiment of **16C**. In this embodiment, gear rack **1710** includes shelves **1714-1** and **1714-2** laterally inward of respective teeth **1716-1** and **1716-2**. Rollers **1712-1** and **1712-2** are secured to an axle **1715** extending through gearbox **1720**. The rollers **1712-1** and **1712-2** ride on the respective shelves **1714-1** and **1714-2**. The force on gears **1740-1** and **1740-2** from spring **1791** is reduced because the force is acting through rollers **1712-1** and **1712-2** on shelves **1714-1** and **1714-2**, thus allowing for easier movement of gears **1740-1** and **1740-2** on teeth **1716-1** and **1716-2**. Also, it is easier to maintain center distance for gear mesh. Similar to FIG. **16B**, handle **1799** can be replaced with manual adjustment **1780**. FIG. **16E** illustrates an alternative embodiment of a depth adjustment assembly **90J-1**, similar to the depth adjustment assembly **90J**, but with rollers **1712-1** and **1712-2** coaxial with gears **1740-1** and **1740-2**. This simplifies the embodiment that is shown in FIGS. **16C** and **16D** to permit the depth adjustment assembly **90J-1** to have a full range of motion across teeth **1716**.

(104) FIG. **17** is a side elevation view of a conventional Case row unit **1810** such as disclosed in U.S. Pat. No. 6,827,029 (the "Case '029 patent"), incorporated herein by reference, which is adapted with another embodiment of a depth adjustment assembly **90X**, as hereinafter described. FIG. **17A** is an enlarged partial view of FIG. **17**. The conventional Case row unit includes an adjustment handle (identified by reference numeral 90 in FIG. 2 of the Case '029 patent) which is removed and replaced with an actuator **1850** coupled to a screw **1841** that engages with the adjustment rod **1860** (corresponding to rod 92 in FIG. 2 of the Case '029 patent). The depth adjustment assembly **90X** is mounted to row unit **1810** via bracket **1870** having bracket arms **1870-1** and **1871-2** attached to channel member **1814**. Actuator **1850** includes motor **1830** and gear box

1820, which drives shaft **1821**, which is coupled to threaded rod **1841** via coupler **1840**. Threaded rod **1841** is threadably engaged with adjustment rod **1860** extending through the channel member **1814**. Adjustment rod **1860** has a receiver end **1861** with a threaded nut **1862** for threadably receiving threaded rod **1841**. Adjustment rod **1860** extends through channel member **1814** and is connected to a rocker **1895** at its distal end. The rocker **1895** is pinned to the distal end of the adjustment rod **1860** and acts on respective gauge wheel arms **1894-1** and **1894-2**. Gauge wheel arms **1894-1** and **1894-2** are pivotally connected to a frame member of the row unit **1810** via pivots **1892-1** and **1892-2**, respectively. Gauge wheels **52-1** and **52-2** are connected to gauge wheel arms **1894-1** and **1894-2**, respectively.

(105) FIGS. **18** and **18A** illustrate another embodiment of a depth adjustment assembly **90K** utilizing a gear rack **1710** and a distance sensor **1717** to determine the position of the actuator **1750B** along the gear rack **1710**. FIG. **18A** is a rear elevation view of FIG. **18**. In this embodiment, the distance sensor **1717** is disposed on the bottom of gear box **1720** and is disposed over a ledge **1721** disposed on an interior surface **1722** of gear rack **1710**. In this embodiment, ledge **1721** has a constantly changing distance with respect to the constant radius of teeth **1716**. Sensing this change in distance, distance sensor **1717** communicates with depth control and soil monitoring system **300** (discussed later).

(106) FIGS. **19** and **19A** illustrate another embodiment of a depth adjustment assembly **90L** utilizing a gear rack **1710** and a distance sensor **1717** to determine the position of the actuator **1750C** along the gear rack **1710**. FIG. **19A** is a rear elevation view of FIG. **19**. In this embodiment, the distance sensor **1717** is disposed on the handle shaft **1798**. The interior wall **1718** of ledge **1723** adjacent to the distance sensor **1717** has a constantly changing width transverse to the direction of travel of handle shaft **1798**. The change in distance to the interior wall **1718** is sensed by the distance sensor **1717** which communicates with depth control and soil monitoring system **300**.

(107) Distance sensor **1717** can be any sensor that can measure distance. Examples of distance sensors include, but are not limited to Hall effect sensors and inductive sensors.

(108) FIGS. **20A** to **20K** illustrate another embodiment of a depth adjustment assembly **90M** utilizing a gear rack **1910** and a distance sensor **1917** to determine the position of actuator **1950** along the gear rack **1910**. In this embodiment, the distance sensor **1917** is disposed above ledge **1921** of gear rack **1910**. In one embodiment, distance sensor **1917** is attached to gear box **1920**. In this embodiment, ledge **1921** has a constantly changing distance with respect to the constant radius of teeth **1916**. Sensing this change in distance, distance sensor **1917** communicates with depth control and soil monitoring system **300**. Alternatively, gear rack **1910** can have an interior wall similar to interior wall **1718** on gear rack **1710** with distance sensor disposed to sense the change in distance to the interior wall (not shown).

(109) In depth adjustment assembly **90M** the actuator **1950** is disposed on and engaged with gear rack **1910**. Actuator **1950** includes an electric motor **1930** connected to and driving gear box **1920**. Gear box **1920** drives gears **1940-1** and **1940-2**. Gears **1940-1** and **1940-2** have teeth **1941-1** and **1941-2**, respectively, for engaging teeth **1916** (**1916-1** and **1916-2**) on gear rack **1910**.

(110) As best viewed in FIG. **20F**, gear box **1920** is connected via shaft **1998** to depth adjustment body **1994** which pivots about pivot **92** to adjust rocker **95**. In one embodiment, shaft **1998** is connected to gear box **1920** via connection **1922** (FIG. **20E**). Shaft **1998** terminates with annular lip **1995** (FIG. **20F**) inside depth adjustment body **1994**. Disposed in depth adjustment body **1994** is force member **1991** (such as a spring) to force shaft **1998** via annular lip **1995** away from depth adjustment body **1994**. In the embodiment when force member **1991** is a spring, annular lip **1995** can have a nub **1997** and depth adjustment body **1994** can have a nub **1996** about which spring **1991** is disposed to help retain spring **1991** within depth adjustment body **1994**.

(111) As best viewed in FIG. **20G**, gear rack **1910** in one embodiment can have one or more protrusions **1929** which may engage with the depth adjustment notches **97** on frame **14**, which are typically found on most frames (not shown).

(112) Gear box **1920** includes wheels **1913-1** and **1913-2** (see FIG. **20H** with the gear box **1920** removed for clarity) attached to its side. Wheels **1913-1** and **1913-2** engage shelves **1919-1** and **1919-2**, respectively, on gear rack **1910**. The engagement of wheels **1913-1** and **1913-2** can be best viewed in FIGS. **20H** and **20I**. FIG. **20I** is a perspective view of gear rack **1710** showing the changing radius of ledge **1921** with respect to teeth **1916-2** and ledge **1919-2**.

(113) FIG. **20J** shows gear box **1920**, and FIG. **20K** shows the internal parts of gear box **1920** with gear box housing **1925** removed to show worm gear **1927**, wheel **1928** (or **1928-1** and **1928-2**) and shaft **1926**. Worm gear **1927** coupled to a rotatable shaft **1931** driven by motor **1930** turns wheel gear **1928** and shaft **1926**. Gears **1940-1** and **1940-2** are disposed about shaft **1926**. In one embodiment, worm gear **1927** and wheel gear **1928** are made from powdered metal. In one embodiment for ease of assembly, wheel gear **1928** is made in two parts, left wheel gear **1928-1** and right wheel gear **1928-2**, all of which can be made from powdered metal.

(114) FIG. **21** is a side elevation view of another embodiment of a depth adjustment assembly **90N**. Assembly **90N** is an alternative to assembly **90M** in which the wheel gears/pinions are replaced with one or more worm gears. In this embodiment, gear box **1980** is connected via shaft **1998** to depth adjustment body **1994** which pivots about pivot **92** to adjust rocker **95**. Disposed on either or both sides of gear box **1980** and positioned over gear rack **1910** is a worm gear **1981** having flights **1982** that engage with the teeth **1916** of the gear rack **1910**. The worm gear **1981** has a shaft **1983** which is rotatably driven by an electric motor **1984**. The shaft **1983** is supported within a U-shaped bracket **1985** which is supported by the gear box **1980**. For consistency with the previously described embodiments, it should be appreciated that the depth adjustment assembly **90N** may comprise corresponding left and right worm gears **1981**, flights **1982**, shafts **1983**, motors **1984** and brackets **1985** differentiated by the suffix “-1” and “-2” for those components disposed over the respective left and right gear teeth **1916-1**, **1916-2** of the gear rack **1910**. However, because FIG. **21** is a sided elevation view, only the “-2” components are visible.

(115) FIGS. **22-26** illustrate another embodiment of a depth adjustment assembly **90P**. As with the previously described embodiment **90M**, the depth adjustment assembly **90P** utilizes an actuator **2050** comprising an electric motor **2030**. The gear box, gear rack, and depth adjustment body and the interaction of the respective components are the same for this embodiment **90P** as was previously described in connection with embodiment **90M**. As such, the same reference numbers are used for the same or corresponding parts in the drawing figures illustrating embodiment **90P** as was used for the embodiment **90M**. Accordingly, in the interest of brevity, since the gear box **1920**, gear rack **1910**, and depth adjustment body **1994** are the same for the two embodiments **90P** and **90M**, the description, operation and interaction of the various components will not be repeated here.

(116) Unlike embodiment **90M**, however, the embodiment of **90P** utilizes a different method of sensing position of the depth adjustment body **1994** with respect to the gear rack **1910**. Referring to FIGS. **24**, **25** and **26**, a ring magnet **2010** and a Hall effect sensor **2012** are used to count the motor revolutions to detect position of the depth adjustment body **1994** with respect to the gear rack **1910**. An accelerometer **2014** also measures the angle (i.e., position) of the depth adjustment body **1994** with respect to the gear rack **1910**. Current sensors **2016**, **2018** detect both directions of rotation of the motor **2030** or the motor shaft **1931** having a worm gear **1927** which engages with the wheel gear or gears **1928** as illustrated in FIG. **20K** and as previously described in connection with embodiment **90M**. The ring magnet **2010** is disposed to rotate with the motor **2030** or motor shaft **1931**. The Hall effect sensor **2012**, accelerometer **2014** and current sensors **2016**, **2018** are mounted to a circuit board **2020** within a protective housing **2022** and are in data communication with the planter monitor **50** discussed later.

(117) Optionally, manual buttons **2023** and **2024** can be mounted to circuit board **2020** for manually advancing depth adjustment assembly **90P** in either direction. Each manual button **2023** and **2024** can be configured to move depth adjustment assembly **90P** in a direction opposite to the

other manual button **2023** or **2024**. Each manual button **2023** and **2024** can be configured to move depth adjustment assembly **90P** a predetermined distance with a push of the button. For example, a push could advance depth adjustment assembly **90P** by $\frac{1}{8}$ inch (0.32 cm). Also, holding either manual button **2023** or **2024** will continuously advance depth adjustment assembly Manual adjustment can be used to manually calibrate depth adjustment assembly **90P** or to move depth adjustment assembly **90P** to a location out of the way to allow an operator to work on any piece of equipment. In another embodiment, pushing both manual buttons **2023** and **2024** can be programmed to execute a selected function, such as moving depth adjustment assembly **90P** to the shallowest or deepest setting.

(118) Calibration

(119) In one method of calibrating the depth adjustment assembly **90P**, the actuator **2050** is “homed” or “zeroed” with respect to a starting position. The zero or home position may be associated with the shallowest setting, with the deepest setting, with the shallowest and the deepest setting, or with a known depth between the shallowest and deepest settings, such as, for example, a two inch depth.

(120) By way of example, to zero or home the actuator **2050** to the shallowest depth, the motor **2030** is commanded to actuate via the planter monitor **50** to cause the gears to engage and rotate to move the depth adjustment body **1994** along the gear rack **2010** toward the shallowest position until the gear box **1920** or depth adjustment body **1994** abuts with a positive stop on the gear rack **1910**. When the positive stop is reached, the current detected by one of the current sensors **2016**, **2018** will begin to spike, indicating the gear box **1920** or depth adjustment body **1994** is at the shallowest setting possible, which establishes the zero or home position. The detected current spike associated with the zero or home position may be between 5% to 20% of the full motor current. For example, if the motor **2030** is an 18 amp motor, when the current reaches 2 amps (i.e., approximately 10% of the full motor current), the motor is commanded to shut off, and that position is associated with the “zero” or “home” position. In other words the motor may be commanded to shut off before the current reaches full current to save the motor from being overworked. In other embodiments, the current spike may be no more than 95% of the full motor current, or no more than 90%, or no more than 80%, or no more than 70%, or no more than 60%, or no more than 50%, or no more than 40%, or no more than 30%, or no more than 25% of the full motor current.

(121) Once home or zero is established, a block of a known thickness (e.g., two inches) is placed under each gauge wheel **52-1**, **52-2** to simulate a known trench depth setting (i.e., the permitted travel distance of the gauge wheels with respect to the opener disks). The motor **2030** is then commanded to actuate to cause the gears to engage and rotate to move the depth adjustment body **1994** toward the deepest trench depth setting until another spike in the current is detected by one of the current sensors **2016**, **2018**, indicating that the gauge wheels are pressing firmly against the blocks and establishing that the predefined trench depth setting has been reached (e.g., the two inch depth corresponding to the block thickness). As the depth adjustment moves from the shallowest position until reaching the predefined block depth setting, the number of revolutions of the motor **2030** or motor shaft **1931** are counted via the Hall effect sensor detecting the rotation of the magnet **2010** (which rotates with the motor **2030** or motor shaft **1931**). The counted number of revolutions, whether of the motor **2030** or motor shaft **1931**, are hereinafter referred to as “revolution counts.” The number of revolution counts will therefore correspond to the predefined depth setting (which, in this example, is a two inch depth based on the known thickness of the block placed under the gauge wheels). Therefore after calibrating to a known block thickness, the relationship (linear or non-linear depending on the shape of the row unit **10**) between revolution counts versus change in depth can be applied to determine the depth throughout the entire range of movement of the actuator **2050** or depth adjustment body **1994** with respect to the gear rack **1910** based on revolution counts.

(122) It should also be appreciated that rather than zeroing or homing the actuator to the shallowest depth, the actuator may be zeroed or homed to the deepest depth. In such a method, the motor **2030** is commanded to actuate via the planter monitor **50** to cause the gears to engage and rotate to move the depth adjustment body **1994** along the gear rack **2010** toward the deepest position until the gear box **1920** or depth adjustment body **1994** abuts with a positive stop on the gear rack **1910**. When the positive stop is reached, the current detected by one of the current sensors **2016**, **2018** will begin to spike, indicating the gear box **1920** or depth adjustment body **1994** is at the deepest setting possible, which establishes the zero or home position. Once zero or home is determined, a block of a known thickness (e.g., two inches) is placed under each gauge wheel **52-1**, **52-2** to simulate a known trench depth setting (i.e., the permitted travel distance of the gauge wheels with respect to the opener disks). A benefit to going to the deepest setting possible is that the weight of the planter is carried by opening discs **62** on each row unit **10** so that the planter does not need to be raised. This allows the block to be placed under each gauge wheel **52**. The motor **2030** is then commanded to actuate to cause the gears to engage and rotate to move the depth adjustment body **1994** toward the shallowest trench depth setting until another spike in the current is detected by one of the current sensors **2016**, **2018**, indicating that the gauge wheels are pressing firmly against the blocks indicating that the predefined trench depth setting has been reached (e.g., the two inch depth corresponding to the block thickness). As the depth adjustment body **1994** moves from the deepest position until reaching the predefined block depth setting, the number of revolutions of the motor **2030** or motor shaft **1931** are counted via the Hall effect sensor detecting the rotation of the magnet **2010** (which rotates with the motor **2030** or motor shaft **1931**). The number of revolution counts will therefore correspond to the predefined depth setting (which, in this example, is a two inch depth based on the known thickness of the block placed under the gauge wheels). Therefore, after calibrating to a known block thickness, the relationship (linear or non-linear depending on the shape of the row unit **10**) between motor revolutions versus change in depth can be applied to determine the depth throughout the entire range of movement of the actuator **2050** or depth adjustment body **1994** with respect to the gear rack **1910** based on revolution counts. After calibration, the depth can be set to maximum again so that the blocks can be removed without the need to raise the planter.

(123) It should also be appreciated that the zero or home position need not be at the shallowest or deepest position. Instead, the zero or home position may be associated with a known depth between the shallowest and deepest settings (e.g., a two inch depth). By way of example, if it is desired to associate the zero or home position with a two inch trench depth (i.e., the permitted travel distance of the gauge wheels with respect to the opener disks), the actuator **2050** or depth adjustment body **1994** may initially be placed at the shallowest or deepest setting and then a two inch thick block may be placed under each gauge wheel **52-1** and **52-2**. The motor **2030** may then be commanded to cause the depth adjustment body **1994** to travel toward the shallowest or deepest position. As the depth adjustment body **1994** moves toward the shallowest or deepest position, the motor revolutions are counted until a current spike is observed as the gauge wheels **52-1**, **52-2** start to contact the block. The position of the depth adjustment body **1994** where the current spike is detected (i.e., at the two inch depth in this example) is established as the zero or home position. When the actuator **2050** or depth adjustment body **1994** is moved from the established zero or home position toward a deeper or shallower setting, the number of rotations of the motor **2030** or motor shaft **1931** (in the direction of the increase or decrease of the depth) are counted from the established zero or home position. As previously described, the linear relationship between revolution counts versus change in depth can be applied to determine the depth throughout the entire range of movement of the actuator **2050** or depth adjustment body **1994** based on revolution counts.

(124) The above described calibration processes may be repeated for each row unit of the planter, which may be performed sequentially for each row unit or all row units may be calibrated

simultaneously. It should be appreciated that only the zero or home position needs to be set, because the thickness of the block used under the gauge wheels is known which sets the trench depth. It should also be appreciated that not all rows need to be zeroed using the above methods. A subset of rows, for example, 50%, 33%, 25%, 20%, or any number less than 100%, may be zeroed or homed according to any of the above methods. The subset of rows that are zeroed are then averaged to provide an average zero or home that can then be applied across all rows equally or to the rows that were not zeroed.

(125) During the calibration process in any of the above examples for detecting the current spikes when the gauge wheels contact the blocks, a downforce sensor may be utilized to ensure consistent loading on the gauge wheels **52-1** and **52-2** across the various row units **10**. For example, a downforce system, such as the DeltaForce® system available from Precision Planting LLC, 23207 Townline Road, Tremont, IL 61568 (described in International Publication No. WO2014/018716), may be utilized to ensure consistent loading on the gauge wheels **52-1** and **52-2** at each row unit **10** to ensure uniform results across the agricultural implement.

(126) It should also be appreciated that the depth adjustment assembly **90P** may be manually calibrated by moving depth adjustment assembly **90P** to a zero or home location by activating manual button **2023** or **2024** to move depth adjustment assembly **90P** as previously described.

(127) When operating in a field, and in particular a no-till field, the gear rack **1910** could become filled with debris, which could limit the travel of the actuator **2050** on the gear rack **1910** causing an earlier current spike due to the depth adjustment body **1994** or gear box **1920** abutting the debris filling the gear rack **1910** instead of the depth adjustment body or gear box **1920** abutting the positive stop on the gear rack **1910**. This earlier or premature current spike due to abutment with the debris will result in a false or incorrect home or zero position. Accordingly, it may be desirable to home or zero the actuator **2050** or depth adjustment body **1994** at a midpoint of the gear rack **1910** or at some other point between either end of the gear rack **1910** where debris is unlikely to accumulate so as to avoiding a false or incorrect home or zero position.

(128) FIG. **27** is a rear perspective view of another embodiment of a depth adjustment assembly **90Q**. In this embodiment, the depth adjustment assembly **90Q** is substantially the same as the previously described depth adjustment assembly **90P**, but incorporates a homing system **1970** for homing the actuator **2050** or depth adjustment body **1994** with respect to a position on the gear rack **1910** between the ends of the gear rack **1910**. The homing system **1970** includes a sensor **1971** and a target **1972**. The target **1972** may be a magnet, and the sensor **1971** may be a Hall effect sensor. Alternatively, the sensor **1971** may be an inductive sensor, and the target **1972** may be a metal block detectable by the inductive sensor. The sensor **1971** may be disposed on the gear box housing **1925** and the target **1972** may be disposed on the gear rack **1910**, such as on one edge **1912-1**, **1912-2** of the shelf **1919-1**, **1919-2** at the midpoint of travel along gear rack **1910** or at another point that is not at either end of gear rack **1910**. The sensor **1971** may be connected to circuit board **2020**, which may be in data communication with the monitor **50**. Home or zero is established when the sensor **1971** is disposed over the target **1972**. Thus, if the target **1972** is at the midpoint of the gear rack **1910**, then the home or zero will be the midpoint of gear rack **1910**. As previously described, the linear relationship between revolution counts versus change in depth can be applied to determine the depth throughout the entire range of movement of the actuator **2050** or depth adjustment body **1994** based on revolution counts in either direction from the zero or home position.

(129) It should be appreciated that the homing system **1970** may be used in conjunction with or in the alternative to the zeroing or homing methods described above.

(130) As a verification or check of the depth based on the number of revolution counts as described above, the revolution counts may be correlated with the angle of the actuator **2050** or the depth adjustment body **1994** as detected by the accelerometer **2014**. The correlation of the revolution counts with actual measured angles detected by the accelerometer **2014** will ensure proper depth

calibration to account for variations in geometry of components from the manufacturer and to account for worn components on the row unit. A standard calibration equation may be used to correlate the depth setting (based on revolution counts) since there is a predetermined number of pulses from the magnet and Hall effect sensor that correlate to a predetermined depth change. For example, tests have shown that 250 revolution counts (producing 250 pulses from the magnet rotating past the Hall effect sensor) correlates to roughly a 0.11 inch (0.28 cm) change in furrow depth, and this change in furrow depth in relation to revolution counts is substantially linear across the entire depth range. Therefore, after calibrating to a known block thickness or to a home position using the homing system **1970**, the linear relationship between revolution counts versus change in depth can be applied to determine the depth throughout the entire range of movement of the actuator **2050** or depth adjustment body **1994**. Similarly, if the change in planting depth is substantially linear in relation to revolution counts, the angle of the actuator **2050** or depth adjustment body **1994** as detected by the accelerometer **2014** can likewise be correlated using a standard calibration equation.

(131) In addition, it is known that different soils, tillage practices and field conditions can affect actual trench depth in comparison to trench depth settings. Thus, for example, if the commanded depth is 2 inches (5 cm), but measurement of the actual depth in the field reveals that the seed trench is only 1.5 inches (3.8 cm), an offset to the calibration may be applied to “true-up” the position of the actuator **2050** or the depth adjustment body **1994** with the actual trench depth to account for field conditions.

(132) It should be appreciated, however, that if the relationship between revolution counts with respect to change in depth across the entire range of movement of the actuator **2050** or depth adjustment body **1994** is not linear, a non-linear relationship may be required to correlate the depth with the revolution counts and with the measured angles detected by the accelerometer **2014**. Such non-linear correlation would be understood by those of ordinary skill in the art.

(133) For any of the depth adjustment assemblies described herein that have a motor as part of their actuator (e.g., **1450, 1550, 1650, 1750, 1850, 1950, 2050**), the set depth can be determined by the motors **1450, 1550, 1650, 1730, 1830, 1930, 1984, 2030** based on their rotations in either direction. If motors **1450, 1550, 1650, 1730, 1830, 1930, 1984, 2030** are stepper motors, the number of steps taken in either direction can be tracked by depth control and soil monitoring system **300**.

(134) Diagnostics

(135) The angles detected by the accelerometer **2014** may be used as a diagnostic tool of the Hall effect sensors on the row units by comparing the angles detected for the depth adjustment body **1994** of one row unit with the readings of the accelerometers of the other row units on the planter. For example, if the angle detected by the accelerometer **2014** of one row unit is significantly different from the angles detected by the accelerometers **2014** of other row units, there may be a faulty Hall effect sensor not accurately counting revolutions of the motor **2030** or motor shaft **1931**.

(136) Additionally, if a repair or alteration is made to the row unit and the actuator **2050** is installed in a different position than its original position, the accelerometer **2014** will detect this discrepancy in relation to the reading of the accelerometers **2014** of the other row units and the discrepancy may be displayed to the operator on the planter monitor to notify the operator that an adjustment is necessary.

(137) Also, the SRM accelerometer may be referenced to verify the position of the actuator **2050**. For example, assuming the planter is operating on flat ground and the accelerometer **2014** and the actuator **2050** is at a 30 degree angle on the gear rack **1910**, but then the topography changes to a 10 degree uphill slope, this change in slope would cause the accelerometer **2014** to signal that the actuator **2050** has moved from its commanded position unexpectedly when in reality it is correct, but the topography changed. To avoid such false readings, the signals generated by the accelerometer **2014** are compared against the signals generated by the accelerometer on the SRM. Thus, as the topography changes, the gravity vector from the SRM changes and this gravity vector

can be referenced in relation to the accelerometer **2014**, to confirm that the actuator **2050** did not move.

(138) Additionally, the current sensors **2016**, **2018** can detect if there is a jamming situation in the actuator **2050** if the current spikes when the depth adjustment body **1994** is not near either end of travel on the gear rack **1910** and the Hall effect sensor fails to pick up any pulses indicating that the motor **2030** has stalled.

(139) Additional Depth Adjustment Assembly Embodiments

(140) FIGS. **28-31** illustrate another embodiment of a depth adjustment assembly **90R**. The depth adjustment assembly **90R** comprises an actuator **3050** which includes an electric motor **3030**. The electric motor **3030**, may be a servo motor, a gear motor or a stepper motor. If a gear motor is used, the gear motor may be geared so low that it is effectively self-locking when not being powered to rotate. Alternatively, if the motor **3030** is not self-locking, a brake may be incorporated to prevent rotation of the worm gear **3032**, the gear wheel **3036** or the shaft **3038** (all discussed later). It should be appreciated that unlike the embodiments **90H-90Q** which either move along the depth adjustment slots **97** (e.g., **90H** and **90H-1** of FIGS. **15-15B**) or which move along a gear rack disposed over the depth adjustment slots **97** (e.g., **90I-90Q** of FIGS. **16-16E** and FIGS. **18-27**), in the embodiment of the depth adjustment assembly **90R**, the actuator/motor **3050/3030** does not travel along the row unit frame **14**. Instead, in the embodiment of the depth adjustment assembly **90R**, the actuator/motor **3050/3030** is fixed in place relative to the frame **14** by a mounting assembly **3000**. The mounting assembly **3000** may include a mounting frame **3002** that is secured to the row unit frame **14** (e.g., by bolts not shown) over the depth adjustment slots **97** of the row unit **10**. A motor mount **3004** is rigidly attached to the mounting frame **3002** from which the motor **3030** and the gear box **3020** are supported.

(141) FIG. **29** is the same view as FIG. **28**, but with the frame **14**, gauge wheel arms **54**, and mounting assembly **3000** removed for clarity. In addition, the gear box housing **3006** is also removed to show the structure and arrangement of the internal gears of the gear box **3020**. FIG. **30** is another perspective view of FIG. **29** looking rearward. The motor **3030** is coupled to and drives a worm gear **3032** which includes helical teeth **3033** around a rotatable shaft **3034**. The helical teeth **3033** engage with notches **3035** of a gear wheel **3036**. The gear wheel **3036** is rotationally fixed to a transverse shaft **3038** and the transverse shaft **3038** is rotationally fixed to a clevis **3040**. A link **3042** pivotally connects the clevis **3040** to the depth adjustment body **3044**. The depth adjustment body **3044** is pivotally connected to the row unit frame **14** by pivot **92**. The depth adjustment body **3044** may be the same as the depth adjustment body **1994** disclosed above in connection with embodiment **90M** (see FIGS. **20E** and **20F**), but in this embodiment, there is no need for the shaft **1998** and spring **1991** as described in connection with embodiment **90M**. A rocker **95** is pivotally attached to the lower end of the depth adjustment body **3044**.

(142) FIG. **31** is a side elevation view of FIG. **29**. As illustrated by dashed lines in FIG. **31**, the rotation of the worm gear **3032** by the motor **3030** causes the gear wheel **3036** to rotate due to the engagement of the helical teeth **3033** with the notches **3035** of the gear wheel **3036**. The rotation of the gear wheel **3036** causes the shaft **3038** to rotate, which, in turn, pivots the clevis **3040**. Since the clevis **3040** is connected to the depth adjustment body **3044** by the link **3042**, the depth adjustment body **3044** is forced to pivot around the pivot **92**, thereby moving the rocker **95** as indicated by dashed lines. As previously explained, adjusting the position at which the rocker **95** engages with the gauge wheel arms **54** limits the upward movement of the gauge wheel arms **54**, which, in turn, limits the upward travel of the bottom of the gauge wheels **52** relative to the bottom of the opening discs **62**, thereby setting the depth of penetration of the opening discs **62** into the soil surface and thus, the depth of the seed furrow **3**.

(143) FIGS. **32-35** illustrate another embodiment of a depth adjustment assembly **90S**. Like the previous embodiment **90R**, there is an actuator **4050** which includes an electric motor **4030**. The electric motor **4030**, may be a servo motor, a gear motor or a stepper motor. If a gear motor is used,

the gear motor may be geared so low that it is effectively self-locking when not being powered to rotate. Alternatively, if the motor **4030** is not self-locking, a brake may be incorporated to prevent rotation of the worm gear **4032**, the gear wheel **4036** or shaft **4038** (all discussed later). As in the embodiment **90R**, the actuator/motor **4050/4030** in this embodiment is fixed relative to the frame **14** by a mounting assembly **4000**. The mounting assembly **4000** may include a mounting plate or frame **4002** that is secured to the row unit frame **14** (e.g., by bolts not shown) over the depth adjustment slots **97** of the row unit **10**. A motor mount **4004** is rigidly attached to mounting frame **4002** from which the motor **4030** and the gear box **4020** are supported.

(144) FIG. **33** is the same view as FIG. **32**, but with the frame **14**, gauge wheel arms **54**, and mounting assembly **4000** removed for clarity. In addition, the gear box housing **4006** is also removed to show the structure and arrangement of the internal gears of the gear box **4020**. FIG. **34** is a perspective view of FIG. **33** looking rearward. The motor **4030** is coupled to and drives a worm gear **4032** which includes helical teeth **4033** around a rotatable shaft **4034**. The helical teeth **4033** engage with notches **4035** of a gear wheel **4036**. The gear wheel **4036** is rotationally fixed to a transverse shaft **4038** and the transverse shaft **4038** is rotationally fixed to a clevis **4040**. A depth adjustment body **4044** includes a slot **4043**. A pin **4042** extends through the slot **4043** and pivotally connects the clevis **4040** to the depth adjustment body **4044**. The depth adjustment body **4044** is pivotally connected to the row unit frame **14** by pivot **92**. A rocker **95** is pivotally attached to the lower end of the depth adjustment body **4044**.

(145) FIG. **35** is a side elevation view of FIG. **33**. As illustrated by dashed lines in FIG. **35**, the rotation of the worm gear **4032** by the motor **4030** causes the gear wheel **4036** to rotate due to the engagement of the helical teeth **4033** with the notches **4035** of the gear wheel **4036**. The rotation of the gear wheel **4036** causes the shaft **4038** to rotate, which, in turn, pivots the clevis **4040**. Since the clevis **4040** is connected to the depth adjustment body **4044** by the pin **4042** movable within the slot **4043**, the depth adjustment body **4044** is forced to pivot around the pivot **92**, thereby moving the rocker **95** as indicated by dashed lines. As previously explained, adjusting the position at which the rocker **95** engages with the gauge wheel arms **54** limits the upward movement of the gauge wheel arms **54**, which, in turn, limits the upward travel of the bottom of the gauge wheels **52** relative to the bottom of the opening discs **62**, thereby setting the depth of penetration of the opening discs **62** into the soil surface and thus, the depth of the seed furrow **3**.

(146) FIGS. **36-38** illustrate another embodiment of a depth adjustment assembly **90T**. Like the previous embodiments **90R**, **90S** there is an actuator **5050** which includes an electric motor **5030**. The electric motor **5030**, may be a servo motor, a gear motor or a stepper motor. If a gear motor is used, the gear motor may be geared so low that it is effectively self-locking when not being powered to rotate. Alternatively, if the motor **5030** is not self-locking, a brake may be incorporated to prevent rotation of the shaft **5034**, the cable drum **5050**, the worm gear **5032**, the gear wheel **5036** or the shaft **5038** (all discussed later). As in the previous embodiments **90R**, the actuator/motor **5050/5030** in this embodiment is fixed relative to the frame **14** by a mounting assembly **5000**. The mounting assembly **5000** may include a motor mount **5004** which also serves to secure the depth adjustment actuator **90T** to the row unit frame **14** (e.g., by bolts not shown) proximate the depth adjustment slots **97** from which the motor **5030** and the gear box **5020** are supported.

(147) FIG. **37** is a perspective view of FIG. **36** looking rearward, but with the frame **14**, gauge wheel arms **54**, and mounting assembly **5000** removed for clarity. In addition, the gear box housing **5006** is also removed to show the structure and arrangement of the internal gears of the gear box **5020**. The motor **5030** is coupled to and drives a worm gear **5032** which includes helical teeth **5033** around a rotatable shaft **5034**. The helical teeth **5033** engage with notches **5035** of a gear wheel **5036**. The gear wheel **5036** is rotationally fixed to a transverse shaft **5038** and the transverse shaft **5028** is rotationally fixed to a cable drum or pulley **5040**. One or more cables **5042** (two are shown) are coiled around the cable drum **5040** with the free ends of the cable **5042** coupled to the depth

adjustment body **5044** by a link **5042**. The depth adjustment body **5044** is pivotally connected to the row unit frame **14** by pivot **92**. A rocker **95** is pivotally attached to the lower end of the depth adjustment body **5044**.

(148) FIG. **38** is a side elevation view of FIG. **36**, but with the frame **14**, gauge wheel arms **54**, mounting assembly **5000** and gear box housing **5006** removed for clarity. As illustrated by dashed lines in FIG. **38**, the rotation of the worm gear **5032** by the motor **5030** causes the gear wheel **5036** to rotate due to the engagement of the helical teeth **5033** with the notches **5035** of the gear wheel **5036**. The rotation of the gear wheel **5036** causes the shaft **5038** to rotate, which, in turn, rotates the cable spool **5040** rigidly attached for rotation therewith, causing the cables **5042** to coil or uncoil from the cable spool **5040** depending on the rotation of the shaft **5038**. Since the depth adjustment body **3044** pivotally attached to the frame **14** by the pivot **92**, as the cables **5042** are coiled or uncoiled from the cable spool **5040**, the depth adjustment body **5044** pivots about the pivot **92** thereby moving the rocker **95** as indicated by dashed lines. As previously explained, adjusting the position at which the rocker **95** engages with the gauge wheel arms **54** limits the upward movement of the gauge wheel arms **54**, which, in turn, limits the upward travel of the bottom of the gauge wheels **52** relative to the bottom of the opening discs **62**, thereby setting the depth of penetration of the opening discs **62** into the soil surface and thus, the depth of the seed furrow **3**.

(149) FIGS. **39-41** illustrate another embodiment of a depth adjustment assembly **90U**. Like the previous embodiments **90R**, **90S**, **90T** the actuator **6050** includes an electric motor **6030**. The electric motor **6030**, may be a servo motor, a gear motor or a stepper motor. If a gear motor is used, the gear motor may be geared so low that it is effectively self-locking when not being powered to rotate. Alternatively, if the motor **6030** is not self-locking, a brake may be incorporated to prevent rotation of the worm gear **6032**, gear wheel **6036** or the shaft **6038** (all discussed later). As in the previous embodiments **90R**, **90S**, **90T**, the actuator/motor **6050/6030** in this embodiment is fixed relative to the row unit frame **14** by a mounting assembly **6000**. The mounting assembly **6000** may include a pair of mounting plates **6002** that are secured to the sides of the row unit frame **14** (e.g., by bolts not shown) proximate the depth adjustment slots **97**. A motor mount **6004** is rigidly attached to the mounting plates **6002** from which the motor **6030** and the gear box **6020** are supported.

(150) FIG. **40** is a perspective view of FIG. **39** looking rearward, but with the frame **14**, gauge wheel arms **54**, mounting assembly **6000** removed for clarity. In addition, the gear box housing **6006** is also removed to show the structure and arrangement of the internal gears of the gear box **6020**. The motor **6030** is coupled to and drives a worm gear **6032** which includes helical teeth **6033** around a rotatable shaft **6034**. The helical teeth **6033** engage with notches **6035** of a gear wheel **6036**. The gear wheel **6036** is rotationally fixed to a transverse shaft **6038** and the transverse shaft **6038** is rotationally fixed to a clevis **6040**. A link **6042** pivotally connects the clevis **6040** to the depth adjustment body **6044**. The depth adjustment body **6044** is pivotally connected to the row unit frame **14** by pivot **92**.

(151) FIG. **41** is a side elevation view of FIG. **39**, but with the frame **14**, gauge wheel arms **54**, mounting assembly **6000** and gear box housing **6006** removed for clarity. As illustrated by dashed lines in FIG. **41**, the rotation of the worm gear **6032** by the motor **6030** causes the gear wheel **6036** to rotate due to the engagement of the helical teeth **6033** with the notches **6035** of the gear wheel **6036**. The rotation of the gear wheel **6036** causes the shaft **6038** to rotate, which, in turn, pivots the clevis **6040**. Since the clevis **6040** is connected to the depth adjustment body **6044** by the link **6042**, the depth adjustment body **6044** is forced to pivot around the pivot **92**, thereby moving the rocker **95** as indicated by dashed lines. As previously explained, adjusting the position at which the rocker **95** engages with the gauge wheel arms **54** limits the upward movement of the gauge wheel arms **54**, which, in turn, limits the upward travel of the bottom of the gauge wheels **52** relative to the bottom of the opening discs **62**, thereby setting the depth of penetration of the opening discs **62** into the soil surface and thus, the depth of the seed furrow **3**.

(152) With respect to each of the depth adjustment assembly embodiments **90R**, **90S**, **90T**, an absolute position sensor may be applied to the respective worm gear **3032**, **4032**, **5032**, **6032**, or to the respective gear wheel **3036**, **4036**, **5036**, **6036** (or to the cable drum **5040** with respect to embodiment **90T**). One such absolute position sensor may be in the form of an inductance sensor that senses a non-concentric curved surface, such that the relative distance changes as the non-concentric surface rotates. One such embodiment of an inductance sensor **7000** is shown in FIGS. **42-44**.

(153) FIG. **42** is a perspective view of an actuator/motor **3050/3030**, **4050/4030**, **5050/5030**, **6050/6030** and gearbox **3020**, **4020**, **5020**, **6020** such as described in connection with each of the respective depth adjustment assemblies **90R**, **90S**, **90T**, **90U**. FIG. **43** is a cross-section view the actuator/motor and gearbox as viewed along lines **43-43** of FIG. **42** showing an embodiment of the inductance sensor **7000**. FIG. **44** is a side elevation view with the gearbox housing **3006**, **4006**, **5006**, **6006** shown in phantom lines for clarity.

(154) The inductance sensor **7000** utilizes contactless inductive sensing to measure the position of a conductive target, an example of a commercially available inductance sensor is the LDC131x or LDC 161x inductance sensor available from Texas Instruments. In a preferred embodiment, two inductive sensors **7002**, **7004** are secured to a sensor board **7006** which extends into the gearbox housing **3006**, **4006**, **5006**, **6006**. The inductive sensors **7002**, **7004** comprise coiled copper with ferrite backing for shielding. The sensor board **7006** is secured to a wall of the gearbox housing **3006**, **4006**, **5006**, **6006** to ensure repeatable and robust placement of the inductive sensors **7002**, **7004** relative to their respective targets **7008**, **7010**. The first inductive sensor **7002** is disposed on the sensor board **7006** relative to the first target **7008** which is secured to a stationary wall or surface of the gearbox housing **3006**, **4006**, **5006**, **6006**. Thus, the first inductive sensor **7002** with its stationary target **7008** serves as a reference sensor to detect expansion or contraction of the gearbox housing that may occur due to temperature variations. The detected changes due to expansion or contraction of the gearbox housing may be used to true-up the measurements of the second inductive sensor **7004**. The second inductive sensor **7004** is disposed on the sensor board **7006** in line with the axis of the transverse shaft **3038**, **4038**, **5038**, **6038** to which the gear wheel **3036**, **4036**, **5036**, **6036** is rotationally fixed. The gear wheel **3036**, **4036**, **5036**, **6036** includes a cam **7020** having a non-concentric or eccentric curved surface **7022** resulting in an eccentric profile which serves as the second target **7010** for the second inductive sensor **7004**. Thus, it should be appreciated that as transverse shaft **3038**, **4038**, **5038**, **6038** is caused to rotate due to engagement of the gear wheel **3036**, **4036**, **5036**, **6036** with the rotating worm gear **3032**, **4032**, **5032**, **6032** rotated by the motor **3030**, **4030**, **5030**, **6030**, the relative distance between the second inductive sensor **7004** and the eccentric profile of the second target **7010** will vary. This change in relative position of the second target **7010** with respect to the stationary second inductive sensor **7004** will thus correspond to the position of the depth adjustment body **3044**, **4044**, **5044**, **6044**, and thus the position of the gauge wheel arms **54** which correspond to the depth of penetration of the opening disc **62**.

(155) It should be appreciated that although the use of the reference sensor **7002** is preferred in order to take into account expansion or contraction of the gearbox housing for purposes of trueing-up the measurements of the second inductance sensor **7004**, the reference sensor **7002** may be omitted such that the inductive sensor **7000** utilizes only one inductive sensor.

(156) Optionally, positive stops **7012**, **7014** are provided within the gear box housing **3006**, **4006**, **5006**, **6006** to engage with the gear wheel **3036**, **4036**, **5036**, **6036** to prevent the gear wheel from rotating too far in either direction, such that the notches **3035**, **4035**, **5035**, **6035** of the gear wheel **3036**, **4036**, **5036**, **6036** remain engaged with helical teeth **3033**, **4033**, **5033**, **6033** of the worm gear **3032**, **4032**, **5032**, **6032** and to prevent the eccentric curved surface **7022** from contact the second inductive sensor **7004**.

(157) The cam **7020** may be removably attached to the gear wheel **3036**, **4036**, **5036**, **6036** such

that different targets **7010** with different with eccentric curved surface profiles **7022** are interchangeable. By providing interchangeable cams **7020** that may be swapped out with one another, the depth adjustment assemblies **90R**, **90S**, **90T**, **90U** and inductance sensor **7000** can be modified depending on the configuration of the gauge wheel arms **54**, the rocker **95** or the depth adjustment body **3044**, **4044**, **5044**, **6044**, which may vary by make or model of the row unit **10**, or to accommodate different applications or uses for the depth adjustment assemblies **90R**, **90S**, **90U**. By way of example, FIGS. **45-47** illustrate one embodiment for removably attaching a cam **7020** to the gear wheel **3036**, **4036**, **5036**, **6036**.

(158) FIG. **45** is an enlarged perspective view showing the cam **7020** attached to the gear wheel **3036**, **4036**, **5036**, **6036**. FIG. **46** is an exploded perspective view of FIG. **45** showing the cam **7020** removed from the gear wheel. FIG. **47** is another perspective view of the cam **7020** shown rotated for illustration purposes. As best viewed in FIGS. **46** and **47**, in one embodiment the gear wheel **3036**, **4036**, **5036**, **6036** includes a lobe **7030** formed or machined on a side of the gear wheel opposite the notches **3035**, **4035**, **5035**, **6035**. The cam **7020** has an inner surface **7024** that mates with lobe **7030** and includes a flange **7026** with an edge **7028** that mates with a corresponding surface of the gear wheel **3036**, **4036**, **5036**, **6036**. The flange **7026** includes apertures **7032** that align with internally threaded bores **7034** in the lobe **7030**. Threaded connectors **7036** extend through the apertures **7032** in the flange and are threadably received by the internally threaded bores **7034** in the lobe **7030**, thus rigidly yet removably securing the cam **7020** to the gear wheel **3036**, **4036**, **5036**, **6036** in a manner that ensures consistent and robust placement of interchangeable targets **7010** to the gear wheel to provide different eccentric curved surface profiles **7022**. Those of skill in the art would recognize and appreciate that there are various other suitable means for removably fastening interchangeable targets **7010** with different eccentric curved surface profiles **7022** onto the gear wheel **3036**, **4036**, **5036**, **6036**, such that the foregoing is but one non-limiting example.

(159) Another absolute position sensor may be in the form of a magnet and a Hall effect sensor, whereby a change in the magnetic field is detected as one rotates relative to the other. For example, a magnet may be attached to the worm gear **3032**, **4032**, **5032**, **6032** with the hall effect sensor secured to a stationary component, such that as the magnet on the worm gear rotates relative to the Hall effect sensor the change in magnetic field is detected and which change in magnetic field is correlated to the position of the depth adjustment body **3044**, **4044**, **5044**, **6044** and thus corresponding to the depth of the furrow **3**.

(160) Additionally, the calibration process and homing system **1970** described above in connection with the depth adjustment assembly **90P** may be used with respect to each of the depth adjustment assembly embodiments **90R**, **90S**, **90T**, **90U**. However, in the embodiments **90S**, **90T**, **90U**, because the actuator **3050**, **4050**, **5050**, **6050** does not move relative to a gear track of the row unit frame **14** as in the embodiment **90P**, the sensor **1971** and a target **1972** comprising the homing system **1970**, would be placed to detect the position of the depth adjustment body **3044**, **4044**, **5044**, **6044** relative to a home or zero position relative to fixed point. For example the sensor **1971** may be mounted on the row unit frame **14** and the target **1972** would be mounted to the depth adjustment body **3044**, **4044**, **5044**, **6044** which pivots about the pivot **92**.

(161) Depth Control Systems

(162) The secondary depth adjustment actuators/motors **720**, **800**, **950**, **1000**, **1230**, **1450**, **1550**, **1650** disclosed herein and the depth adjustment actuators/motors **1650**, **1750/1730**, **1850/1830**, **1950/1930**, **1984**, **2050/2030**, **3050/3030**, **4050/4030**, **5050/5030**, **6050/6030** may be in data communication with a depth control and soil monitoring system **300** as illustrated in FIG. **11** and described herein.

(163) In the depth control and soil monitoring system **300**, a monitor **50** may be in electrical communication with components associated with each row unit **10** including seed meter drives **315**, seed sensors **305**, the GPS receiver **53**, downforce sensors **392**, downforce valves **390**, depth

adjustment actuators **380**, and depth actuator encoders **382** (and in some embodiments actual depth sensors **385** such as those described in Applicant's International Patent Publication No. WO2014/066654, incorporated by reference herein). In some embodiments, particularly those in which each seed meter **30** is not driven by an individual drive **315**, the monitor **50** may also be in electrical communication with clutches **310** configured to selectively operably couple the seed meter **30** to the drive **315**.

(164) Continuing to refer to FIG. **11**, the monitor **50** is in electrical communication with a cellular modem **330** or other component configured to place the monitor **50** in data communication with the Internet, indicated by reference numeral **335**. Via the Internet connection, the monitor **50** receives data from a soil data server **345**. The soil data server **345** may include soil map files (e.g., shape files) associating soil types (or other soil characteristics) with GPS locations, RTK (Real Time Kinematic) data layer, elevation, or topography. In some embodiments, soil map files are stored in the memory of the monitor **50**.

(165) The monitor **50** may also be in electrical communication with one or more temperature sensors **360** mounted to the planter and configured to generate a signal related to the temperature of soil being worked by the planter row units **10**. In some embodiments one or more of the temperature sensors **360** comprise thermocouples disposed to engage the soil as disclosed in Applicant's International Patent Publication No. WO2014/153157, the disclosure of which is incorporated herein in its entirety by reference. In such embodiments, the temperature sensors **360** may engage the soil at the bottom of the trench **38**. In other embodiments, one or more of the temperature sensors **360** may comprise a sensor disposed and configured to measure the temperature of the soil without contacting the soil as disclosed in International Patent Publication No. WO2012/149398, the disclosure of which is hereby incorporated herein in its entirety by reference.

(166) Referring to FIG. **11**, the monitor **50** may be in electrical communication with one or more moisture sensors **350** mounted to the planter and configured to generate a signal related to the temperature of soil being worked by the planter row units **10**. In some embodiments, the moisture sensor **350** comprises a reflectance sensor such as that disclosed in U.S. Pat. No. 8,204,689, hereby incorporated herein in its entirety by reference. In such embodiments, the moisture sensor **350** may be mounted to the shank **15** of the row unit **10** and disposed to measure the soil moisture at the bottom of the trench **38**, preferably at a position longitudinally forward of the seed tube **32**. The monitor **50** may also be in electrical communication with one or more second-depth moisture sensors **352**. The second-depth moisture sensor **352** may comprise a reflectance sensor such as that disclosed in previously referenced U.S. Pat. No. 8,204,689, disposed to measure soil moisture at a depth at which consistent moisture reading is expected. In some embodiments the second-depth moisture sensor **352** is disposed to measure soil moisture at a greater depth than used for planting, such as between 3 and 6 inches and preferably approximately 4 inches below the soil surface. In other embodiments the second-depth moisture sensor **352** may be disposed to measure soil moisture at a lesser depth than used for planting, such as between 0.25 and 1 inch (0.64 and 2.54 cm) and preferably approximately 0.5 inches (1.3 cm) below the soil surface. The second-depth moisture sensor **352** may be disposed to open a trench laterally offset from the trenches **38** opened by the row units **10**.

(167) Referring to FIG. **11**, the monitor **50** may be in electrical communication with one or more electrical conductivity sensors **365**. The electrical conductivity sensors **365** may comprises one or more electrodes disposed to cut into the soil surface such as the sensors disclosed in U.S. Pat. Nos. 5,841,282 and 5,524,560, both of which are hereby incorporated herein in their entirety by reference.

(168) Referring to FIG. **11**, the monitor **50** may also be in electrical communication with one or more pH sensors **355**. In some embodiments the pH sensors **355** are drawn by a tractor or by another implement (e.g., a tillage implement) such that data is stored in the monitor **50** for later use.

In some such embodiments, the pH sensors **355** may be similar to those disclosed in U.S. Pat. No. 6,356,830. In some embodiments, the pH sensors **355** are mounted to the toolbar **8**, preferably at a position laterally offset from the row units **10**.

(169) Depth Control Methods

(170) According to some exemplary processes of controlling depth using the depth adjustment assemblies described herein, a user may manually adjust the primary and/or secondary depth adjustment assemblies.

(171) According to some exemplary processes, the user may manually adjust the primary depth adjustment assembly and may use the monitor **50** to command a depth adjustment to the secondary depth adjustment assembly.

(172) According to some exemplary processes, the user may manually adjust the primary depth adjustment assembly and the monitor **50** may command a desired depth adjustment to the secondary depth adjustment assembly (e.g., one of the actuators/motors **720, 800, 950, 1000, 1230, 1450, 1550, 1650**) by receiving one or more agronomic variables from the sensors (e.g., sensors **350, 355, 360, 365, 352, 385**) or from the soil data server **345** and determine a desired depth adjustment by consulting a database or algorithm relating one or more agronomic variables to a desired furrow depth.

(173) According to some exemplary processes, the monitor **50** may command a desired depth adjustment to the depth adjustment assembly **1650, 1750, 1850, 1950, 1984, 2050, 3050, 4050, 5050, 6050** by receiving one or more agronomic variables from the sensors (e.g., sensors **350, 355, 360, 365, 352, 385**) or from the soil data server **345** and determine a desired depth adjustment by consulting a database or algorithm relating one or more agronomic variables to a desired furrow depth.

(174) According to some exemplary processes, the monitor **50** may command a desired depth adjustment to the depth adjustment assembly **1650, 1750, 1850, 1950, 1984, 2050, 3050, 4050, 5050, 6050** by determining the GPS-reported location of the row unit **10** and consulting a depth prescription map spatially relating locations and/or regions in the field to desired furrow depths.

(175) Depth prescriptions are based on placing seeds at a proper depth to obtain desired germination and emergence. Factors that are used to determine a proper depth include, but are not limited to, soil type, organic matter content, moisture, soil temperature, soil texture, topography, and elevation. The depth prescription can be based on a combination of current temperature and moisture conditions in the field and the predicted temperature and moisture delivery from a weather forecast. This process is described in U.S. Patent Publication No. 2016/0037709, which is incorporated herein by reference.

(176) In another embodiment, a minimum depth and a maximum depth set by an operator may be input into monitor **50** to control the desired depth within a range specified by the operator. The operator set minimum depth may be greater than the actual minimum depth obtainable by the depth adjustment assembly, and the operator set maximum depth may be less than the actual maximum depth obtainable by the depth adjustment assembly. This can be useful to constrain the depth to a desired depth range. For a given seed type, there can be a desired depth range for planting the seed so that the seed can germinate and emerge. When adjusting the depth based on a measured condition in the field, such as moisture, soil temperature, organic matter content, soil type, or soil texture using a sensor, such as described in U.S. Patent Publication No. US2016/0037709, the sensor may signal a depth change to go to a depth that has the selected soil property, but while trying to change to a depth to achieve the selected property, the depth may be outside of the selected range. Having the operator set minimum and maximum depths keeps the seed in the desired depth range. As an example, an operator may want corn seed to be planted between 1.75" and 2.5". If the sensor is measuring for moisture, and the moisture is insufficient at shallower depths, the depth adjustment mechanism would receive a signal to change to a deeper depth, but it can be constrained to stay within the operator selected range.

(177) In some embodiments, the monitor **50** may record changes in depth in the field by associating commanded actuations of the actuator/motor **720, 800, 950, 1000, 1230, 1450, 1550, 1650, 1750/1730, 1850/1830, 1950/1930, 1984, 2050/2030, 3050/3030, 4050/4030, 5050/5030, 6050/6030** with GPS locations reported by the GPS receiver **52**. In some such embodiments, the monitor **50** may record a change in depth concurrently with the commanded actuations of the actuator/motor **720, 800, 950, 1000, 1230, 1450, 1550, 1650, 1750/1730, 1850/1830, 1950/1930, 1984, 2050/2030, 3050/3030, 4050/4030, 5050/5030, 6050/6030**. However, in operation, the force between the rocker **95** and the gauge wheel arm and/or the depth adjustment member may vary, e.g., as the row unit moves across uneven terrain. Thus, in some embodiments, the monitor may monitor the force on the gauge wheel arm and/or the depth adjustment rocker and record the change in depth only when the force is lower than a predetermined threshold. For example, with respect to the embodiment of FIG. **12**, the monitor **50** may monitor the force on the gauge wheel arm and/or the depth adjustment rocker and record the change in depth only when the force is lower than a predetermined threshold at which the depth adjustment member can be advanced for a given position of the actuator **1230**. The force on the gauge wheel arm and/or the depth adjustment rocker may be recorded by a load sensor such as a strain gauge mounted to the gauge wheel arm or other location through which the force is transmitted, or by a load sensing pin incorporated in the row unit as is known in the art.

(178) In other implementations, the monitor **50** may command a temporary change (e.g., reduction) in row unit downforce applied by the downforce actuator **18** concurrently with (or before or after) a commanded change in the extension of the actuator/motor **720, 800, 950, 1000, 1230, 1450, 1550, 1650, 1750/1730, 1850/1830, 1950/1930, 1984, 2050/2030, 3050/3030, 4050/4030, 5050/5030, 6050/6030** in order to permit the depth adjustment. The monitor **50** may then command the row unit downforce applied by the actuator **18** to return to its previously commanded level.

(179) Various modifications to the embodiments and the general principles and features of the apparatus, systems and methods described herein will be readily apparent to those of skill in the art. Thus, the appended claims should not be limited to the embodiments of the apparatus, systems and methods described herein and illustrated in the accompanying drawing figures, but should be accorded the widest scope consistent with their general teachings.

Claims

1. An agricultural row unit, comprising: a row unit frame; a furrow opening disc rotatably supported by said row unit frame for opening a furrow in a soil surface as the row unit frame advances in a forward direction of travel; a gauge wheel disposed adjacent to said furrow opening disc and pivotally supported from said row unit frame by a gauge wheel arm such that said gauge wheel is displaceable with respect to said furrow opening disc; a depth adjustment assembly, including: a depth adjustment body pivotally connected via a pivot to said row unit frame, said depth adjustment body disposed to engage said gauge wheel arm to limit an amount of upward travel of said gauge wheel with respect to said furrow opening disc; a motor operable to cause rotation of a shaft, said shaft operably coupled with said depth adjustment body, whereby rotation of said shaft causes said depth adjustment body to pivotally move about said pivot thereby changing a position of contact of said depth adjustment body with said gauge wheel arm thus changing the amount of upward travel of said gauge wheel with respect to said furrow opening disc wherein said shaft is rotationally fixed to a clevis, said clevis pivotally linked to said depth adjustment body.

2. The agricultural row unit of claim 1, wherein said motor drives a worm gear, and wherein said worm gear is engaged with a gear wheel, and wherein said gear wheel is rotationally fixed to said shaft.

3. The agricultural row unit of claim 2, further comprising: an absolute position sensor disposed to

detect a rotational position of one of said, worm gear and said gear wheel.

4. The agricultural row unit of claim 3, wherein said absolute position sensor is an inductance sensor.

5. The agricultural row unit of claim 4, wherein said absolute position sensor includes a Hall effect sensor and a magnet.

6. The agricultural row unit of claim 4, wherein said inductance sensor includes an inductive sensor disposed relative to a rotating target having a non-concentric curved surface, whereby said inductive sensor senses changes in relative distance to said rotating target.

7. The agricultural row unit of claim 6, wherein said non-concentric curved surface is removably attachable to said gear wheel.

8. The agricultural row unit of claim 2, further comprising a brake configured to prevent rotation of one of said worm gear, said gear wheel and said shaft.

9. The agricultural row unit of claim 1, wherein said motor is an electric motor, and said electric motor is one of a servo motor, a gear motor and a stepper motor.

10. The agricultural row unit of claim 1, wherein said motor is a self-locking motor.

11. The agricultural row unit of claim 10, wherein said self-locking motor is a gear motor that is geared low to effectively self-lock when said geared motor is not powered to rotate.

12. The agricultural row unit of claim 1, further comprising: a rocker connected to said depth adjustment body and engaging said gauge wheel arm.
